Expected goals in soccer

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Introduction

Expected goals(xG) is the most main metric used into the Football Analytics field nowadays. It finds the probability that a shot ends up in a goal.
This report introduces the xG models constructed for seven soccer competitions. The goal for the model as well as this report is to figure out what influence the scoring quality of the team.

Data and methods

Data explanation

The data sets are imported from 7 separated competition record files. Each of the data set contains "period", "time", "x", "y", "freeKick", "rightFoot", "leftFoot", "head", "opportunity" and "goals". These data frames are vertically combined with *pandas.concat*() function in python. Considering the competition level could also affect the scoring results, a new column named "Competition" is created to mark which competition the relevant records belongs to.
The "period" are 1-2 binary variables shows which half the goals or no-goals events are in. The "time" means how long the competition goes in seconds from the half started. "x" and "y" emphasise the location on the shooting. "freeKick" is a 0-1 binary variable showing whether the shooting is a free kick. "rightFoot", "leftFoot", "head" shows the body part used for shooting, they are all binary variables. "opportunity" shows where there is a chance for shooting and "goal" shows whether the shooting is a goal.
Before constructing the model, the correlation matrix between variables is illustrated in the Figure 1. The shooting with different body parts has higher correlation with each other than other variables do.

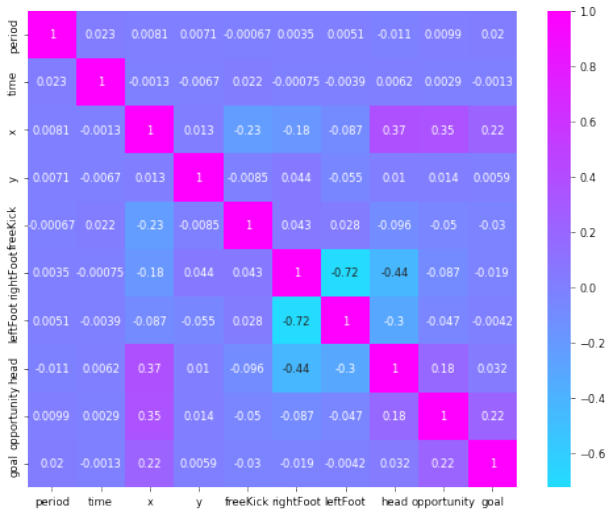

Figure 1: correlations between variables

Data Process and model selection

The scoring probability can be influenced by the level of competition, location for shooting, the body parts used for shooting and the good chance for scoring. Time and period may also influence the shooting.

Considering the length of the goal and the unknown point where the ball finally reached, there are new variables added to the existing data frame, called "view area", "distance up" and "distance down". The "view area" represents the angle range of the shooting location to the two sizes of the goal, and the "distance up" and "distance down" represents the distance to the points.

The analysis initially included all body parts, distances, view area, opportunity, free kick, time and period as the independent variable, and goal as the dependent variable. But after several trials, the period and time are dropped for time efficiency for the final results, because the results shows these variables has very few influence on the dependant variable.

Figure 2 visualizes the successful scoring path and the possibility for scoring. On the left side, the figure shows the scoring path that $goal = 1$ in the sample data.

The "Competition" variables are transformed into binary data type by calling the function $pandas.get_dummies()$ so that these variables could be considered during modelling. The distances and angle (view area), three body parts can be respectively interacted with the other. But since the missing information on the habitual kicking body part(s), and the same angle(distances) can lead to different distances(angles), so these variables were not ignored.

Support vector machine was first chosen for the analysis, however, the results was not satisfying. The Gradient boosting classifier and logistic regression model are then carried out to compare the results. All these models are supervising learning method commonly

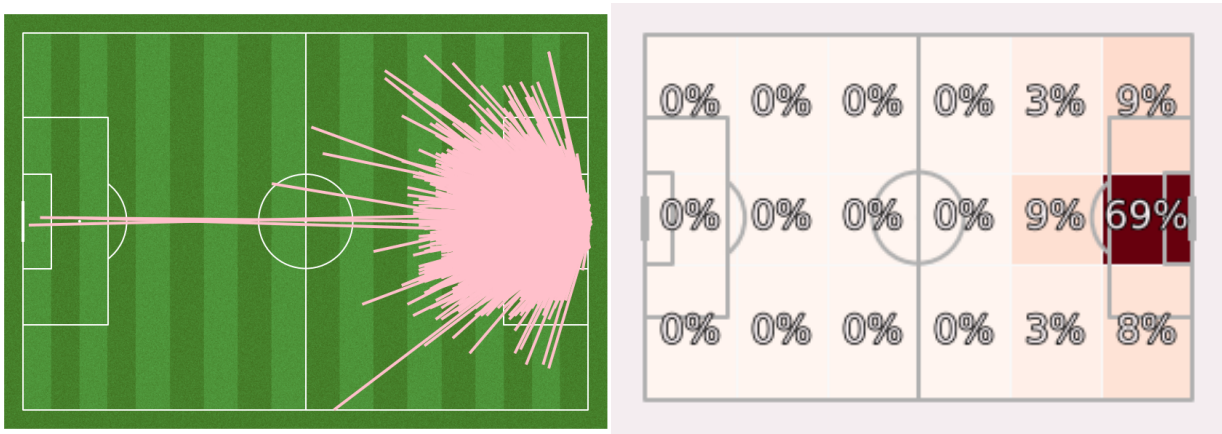

Figure 2: Scoring visualization

used in data classification, but in this case the logistic regression model got the best performance in analyzing the problem. The relative results will be presented in the following section.

Results

The data are randomly split into training data and validation data with a percentage of 70:30. The models were trained with training data and validated with validation data. The measure criteria and the results are shown in table 1.
The logistic regression model has achieved the highest values for recall, f1-score and AUC, which provide the fact that logistic regression model performs the best among the three models.

	Logistic Regression	Support Vector Machine	Gradient Boosting
precision	0.22	0.16	0.43
recall	0.72	0.62	0.18
f1-score	0.34	0.26	0.25
AUC	0.79	0.68	0.77

Table 1: Model evaluation results

The weights for all models are computed, but considering the logistic regression model has the best performance, the weights under this model is more reliable. These weights are shown in table 2
In this xG model, opportunity has the largest influence on shooting. There are also positive relationship for shooting angle, as well as the bias implied in the model to the goal event. The bias may be caused by the interaction between distances and angles, or by

Weights	features
+3.573	opportunity
+1.043	BIAS
+0.698	view area
-0.036	distance up
-0.056	distance down
-1.049	competition France
-1.118	competition Italy
-1.128	competition Spain
-1.174	competition Germany
-1.796	rightFoot
-1.885	competition World Cup
-1.891	leftFoot
-2.028	competition European Championship
−2.730	head

Table 2: Weights for features of logistic regression model

the interaction between shooting by different body parts. The distances, competitions and body parts variables are all negatively related to the goal event. Especially higher level competitions, like World Cup and European Championship, and head have large negative weights, which tells that it is harder to score in higher level competitions or with head. The results seem to be aligned to common sense.

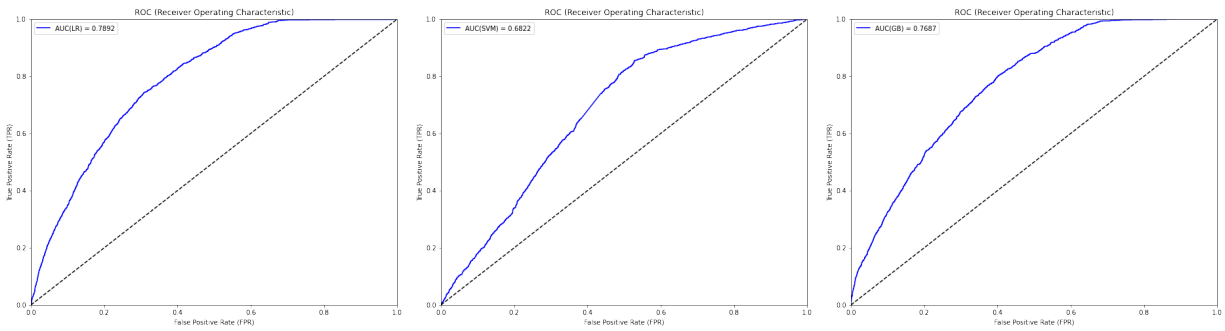

Figure 3: ROC curves

Conclusion and discussion

The results shows that the a good chance for shooting weighed most to successfully score. A good angle also contribute to success shooting. The
However, the models are limited, especially the logistic regression model. The best logistic

regression model was planned to be find by *GridSearchCV* in python, however, some solver did not work well within the short period.
On the data side, when calculating the distance and angles, only ranges can be given. The imprecise calculation for distance and angle could introduce bias, more information should be collected for analysis, for example, the habitual kicking foot of who make the scoring, the ending point of the ball, etc. These information can be obtained by checking the athletes' records or asking athletes. When taking this record, the ending point can also be marked. The distances and angles can be calculated more precise if there are ending points for balls.